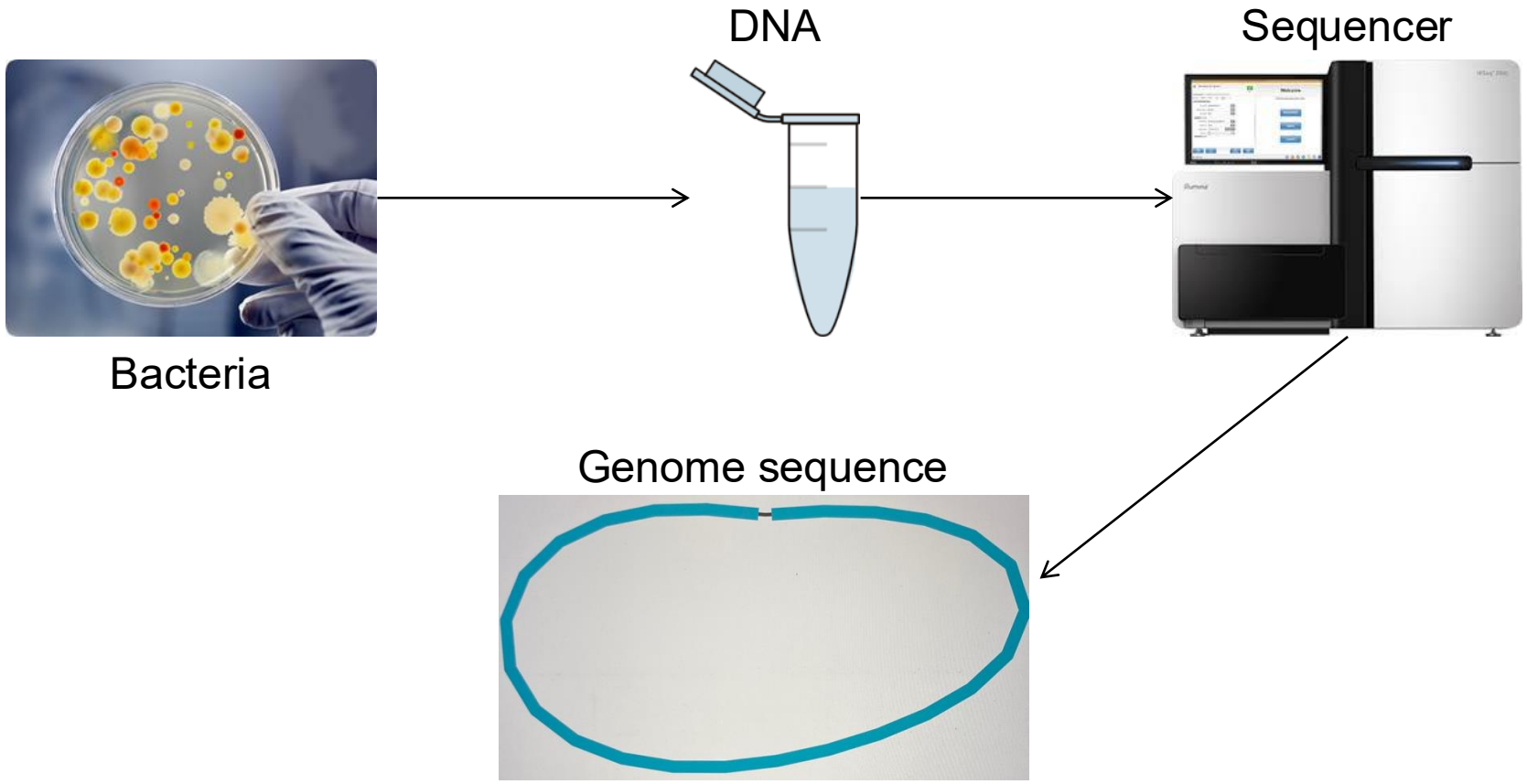
Day 4: Genome Assembly

MMB-127

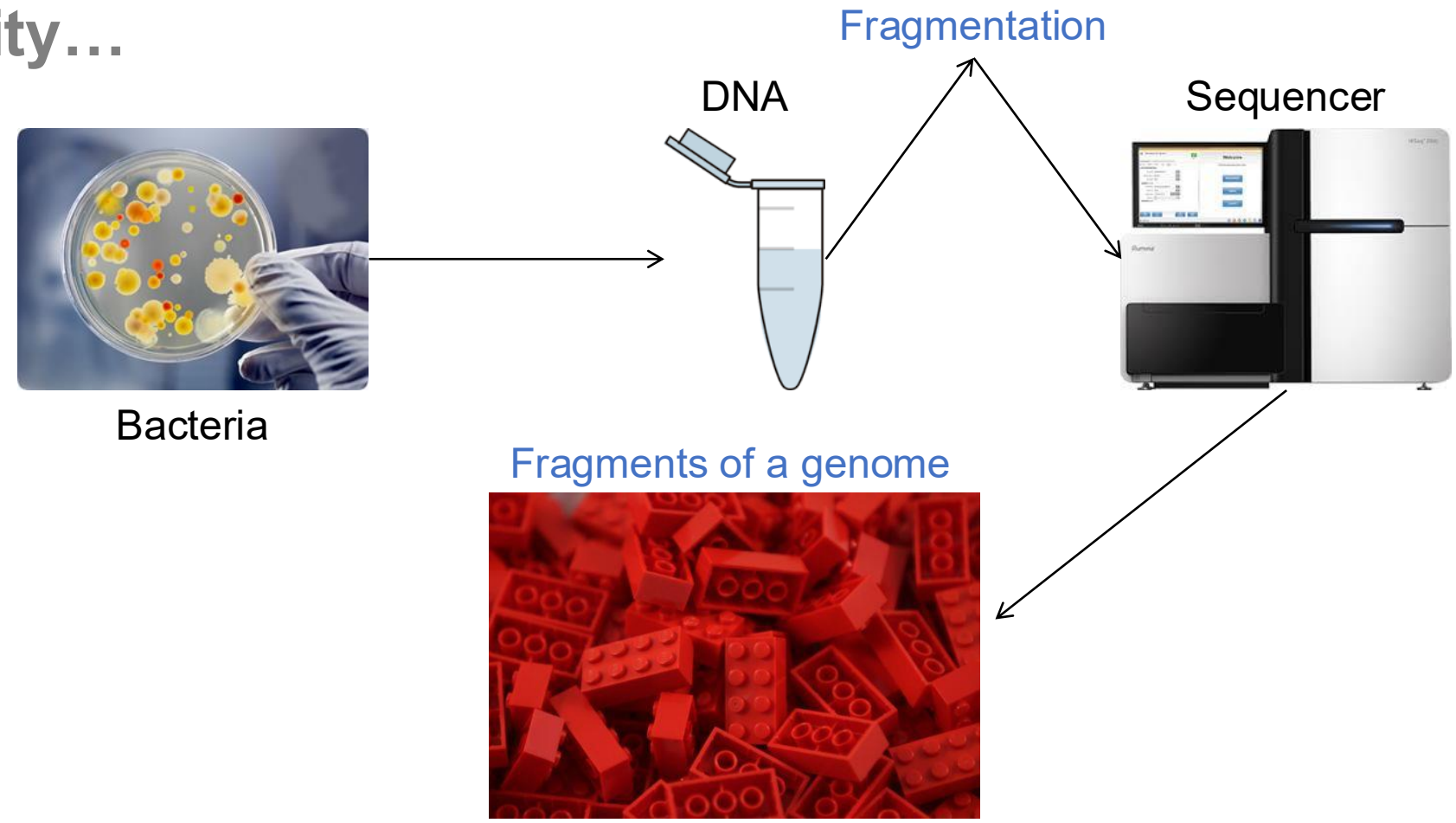
De novo genome assembly

In an ideal world:



De novo genome assembly

In reality…



Genome assembly

- Reconstruct the original genome from long/short sequence reads

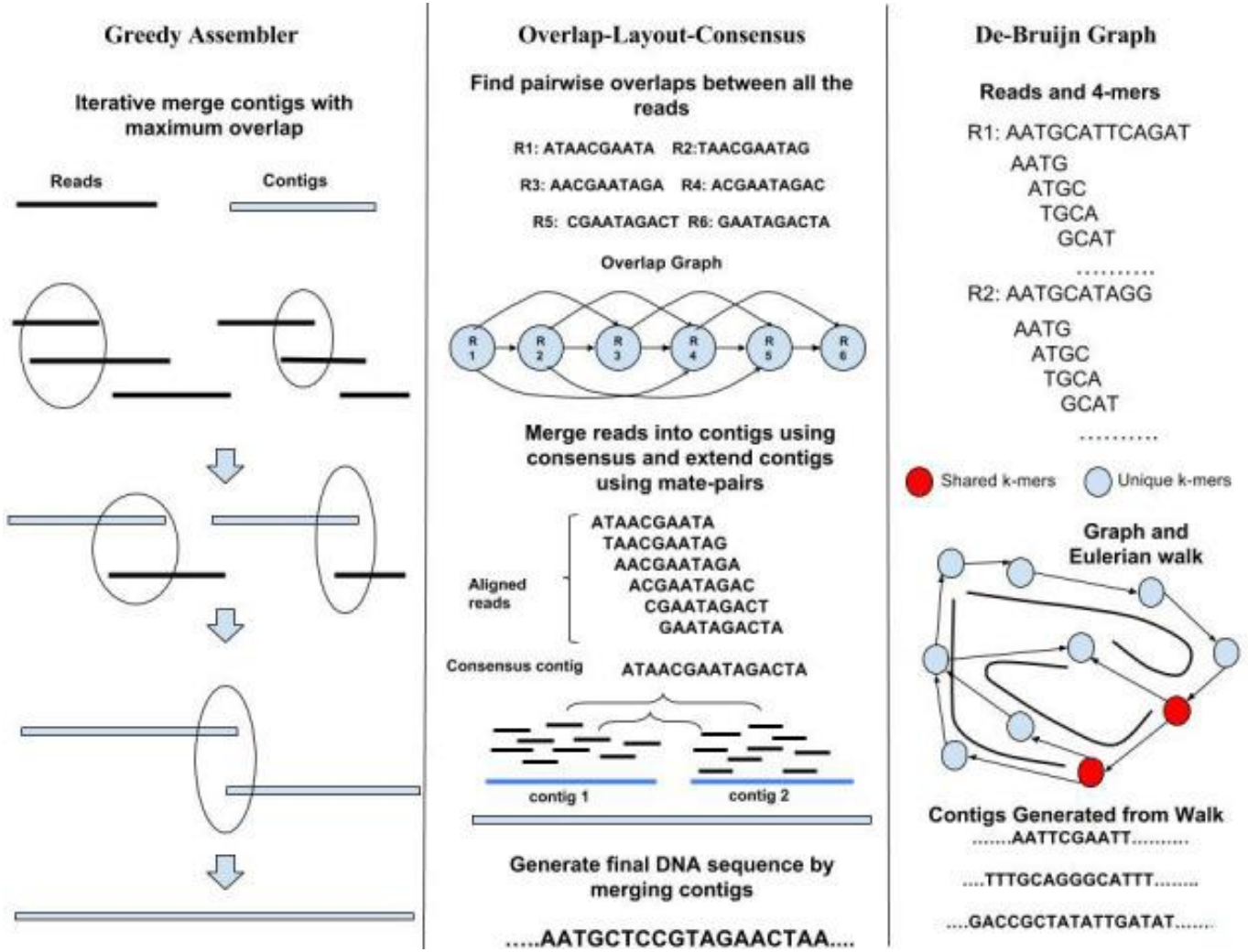
Short reads:

- **In an ideal world:** sequences = one complete genome
- **In reality:** sequences = multiple contigs
  - Contiguous, unambiguous stretches of sequences

Long reads:

- Sometimes we reach the ideal world

Assembly strategies



https://doi.org/10.1093/bib/bbz020

Assemblers

- Bioinformatic tools that combine short sequencing reads into longer contigs
- **Spades:** kmer-based assembler for short reads (accepts also long reads). De Bruijn graphs.
- **Flye:** uses repeat graphs. Can tolerate the higher noise of single-molecule sequencing reads.

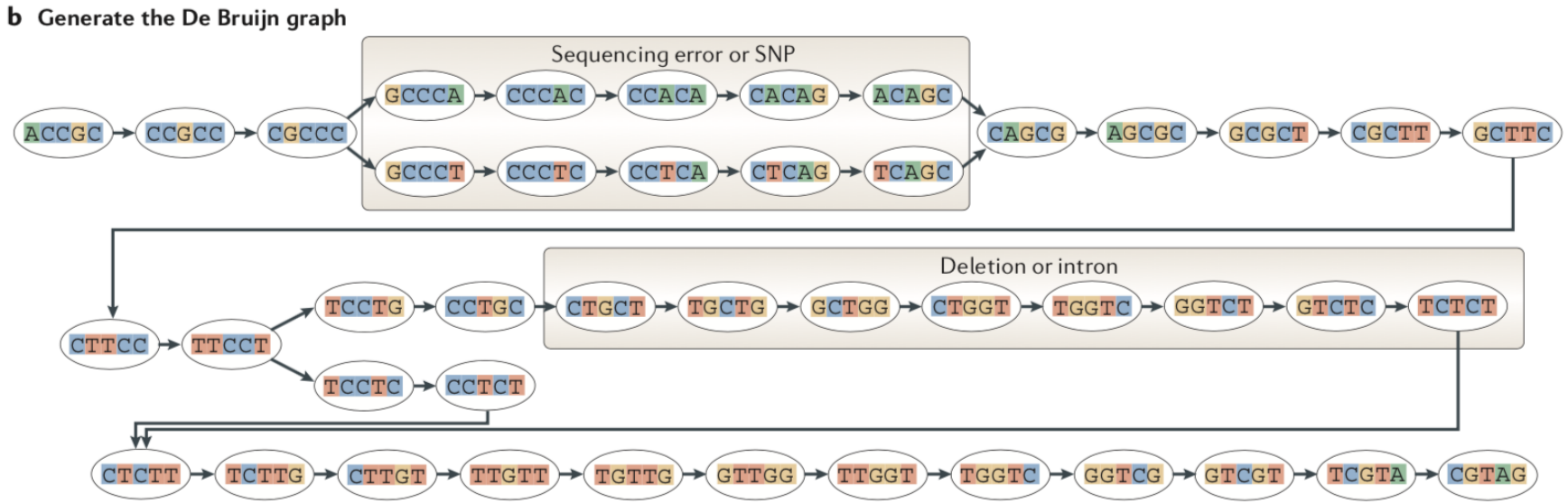
De Bruijn graph

- In real life 10^6 sequences cannot be compared with each other
  - 10^6 x 10^6 = 10^{12} comparisons
- Sequences are reduced to k-mers
  - Smaller subsets of length k



https://doi.org/10.1038/nrg3068

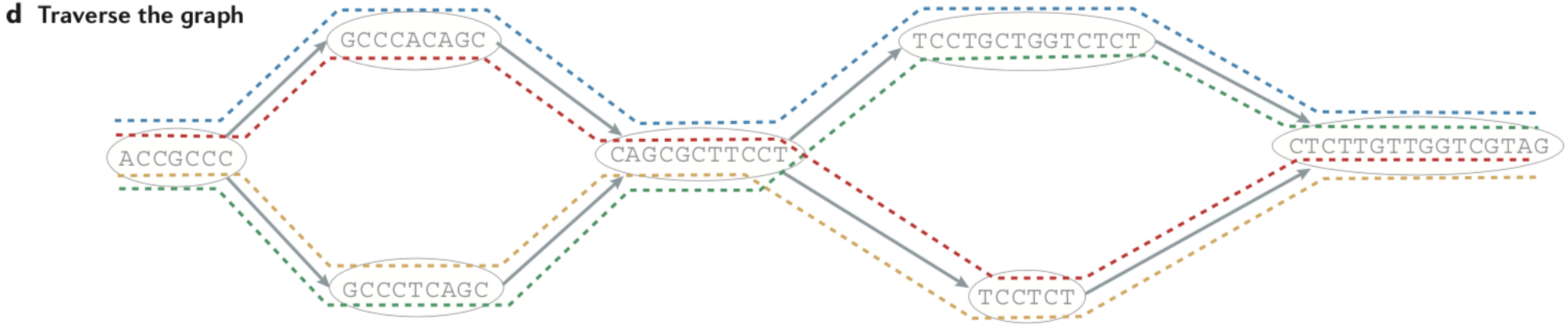
De Bruijn graph





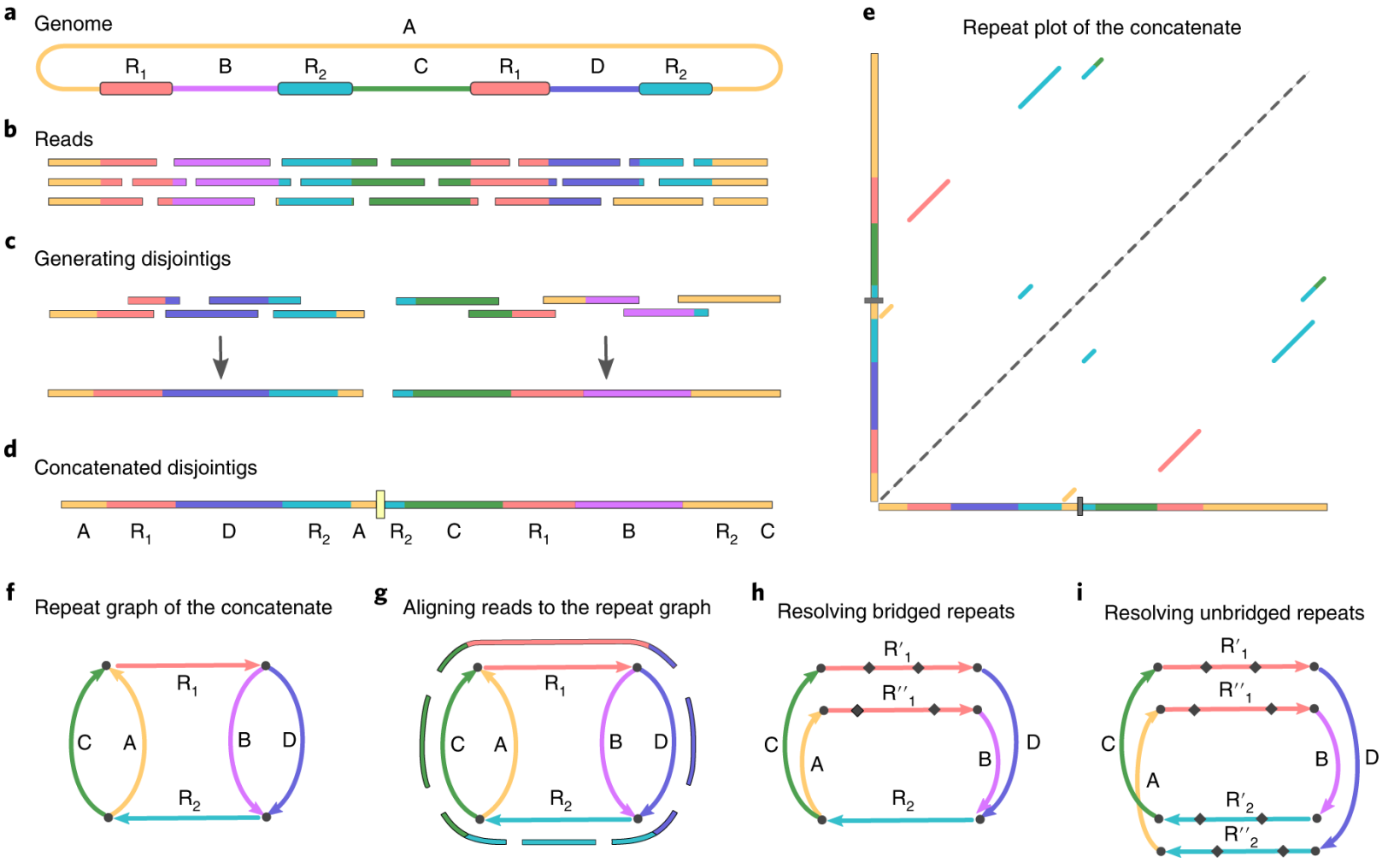
https://doi.org/10.1038/nrg3068

De Bruijn graph



https://doi.org/10.1038/nrg3068

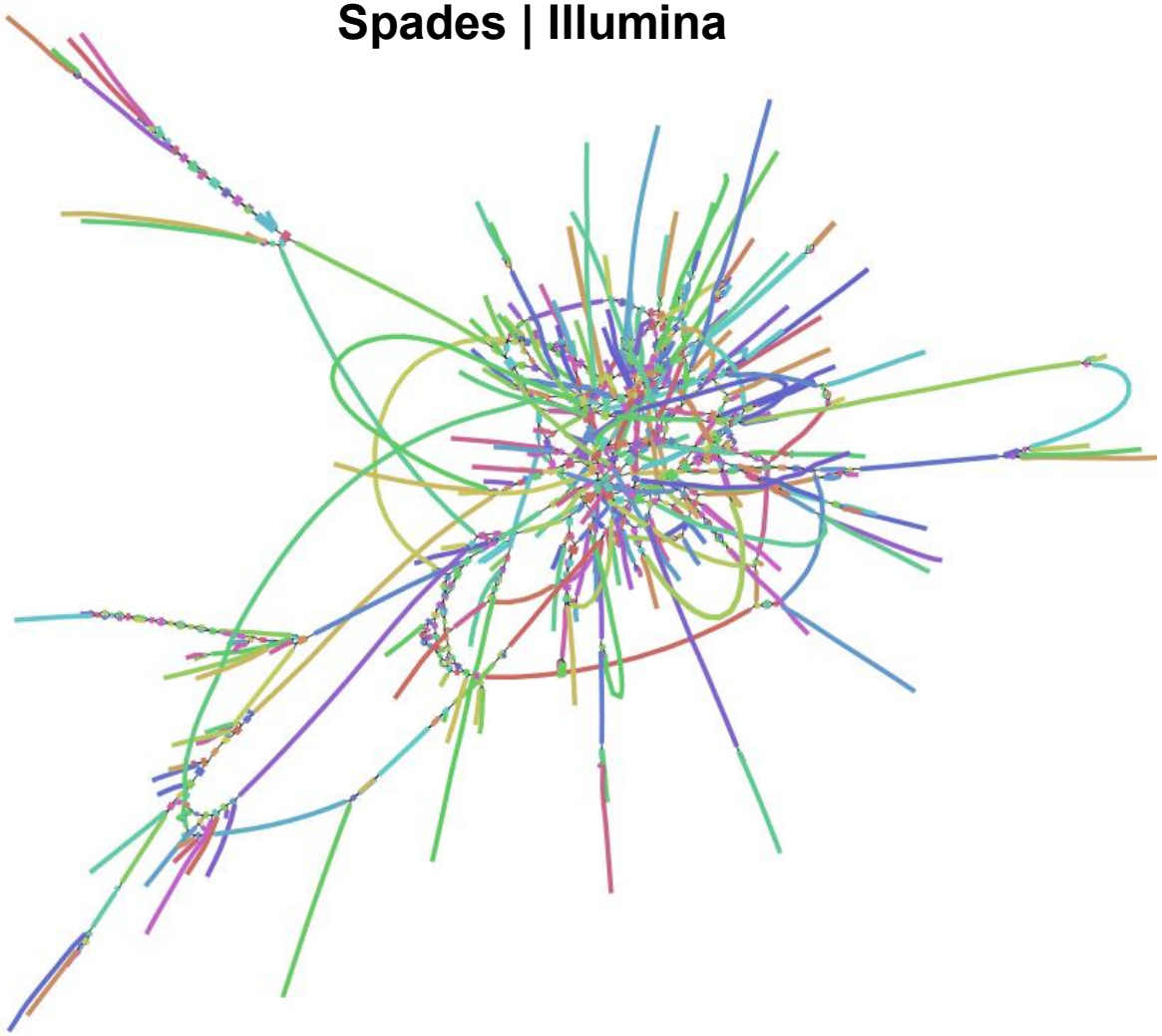
Repeat graphs



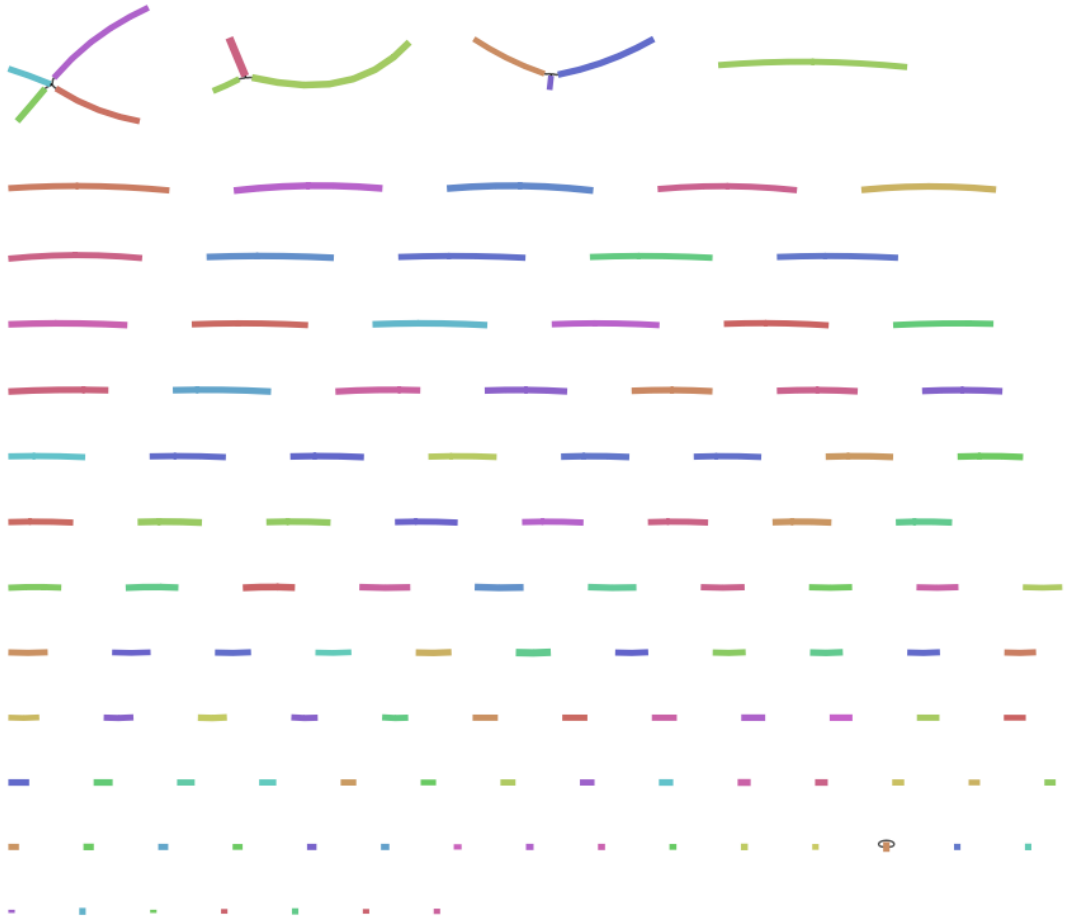
https://doi.org/10.1038/s41587-019-0072-8

Real world assemblies

Spades | Illumina



Flye | Nanopore

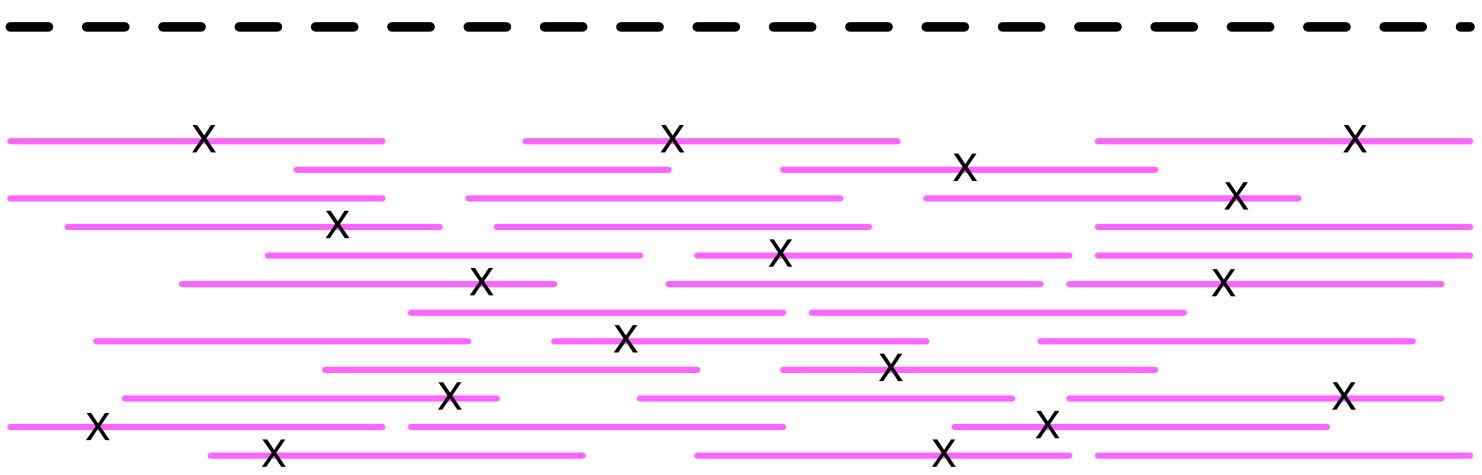


What makes assembly tricky?

- Many pieces (computationally-intensive)
- Errors in sequence (which one is correct?)
- Missing fragments
- Repetitive fragments
- Multiple copies (rRNA gene as an example)
- Circular genome (no starting point)
- Choice of k-mer
  - Too small → misassembly (anything can assemble)
  - Too long → no assembly

Sequencing coverage

- Coverage describes the number of time that each base of the genome is present in the reads
- Assemblers expect equal and high enough coverage (>50x) to work optimally in genome assembly.



Estimated average coverage of your genome

- Your genome size: **X bp**
- Received sequence data: **Y bp**

- Coverage = Y / X

Assembly quality

What is a good assembly?

- Good coverage throughout the contigs
- Correct size (for bacteria, not 400 kb or 14 Mb)
- As few contigs as possible
- Similar GC across contigs

Additional notes about Puhti

Batch jobs:

- sbatch
- scancel
- squeue
- seff

batch_job.sh

```
#!/bin/bash -l
#SBATCH --job-name spades
#SBATCH --output spades_out_%j.txt
#SBATCH --error spades_err_%j.txt
#SBATCH --time 1:00:00
#SBATCH --nodes 1
#SBATCH --ntasks-per-node 1
#SBATCH --cpus-per-task 4
#SBATCH --mem 5000
#SBATCH --account project_XXX

module load spades

spades.py -1 Data/MMB-114_trimmed_1.fastq.gz \
          -2 Data/MMB-114_trimmed_2.fastq.gz \
          -o SPADES \
          -t 4 \
          --careful
```

sbatch batch_job.sh

Assembly outputs

Flye

- `assembly.fasta` – assembled contigs
- `assembly_graph.gfa` – assembly graph
- `assembly_info.txt` – information about each contig

Spades

- `contigs.fasta` – assembled contigs
- `assembly_graph.fastg` – assembly graph

Let’s assemble your genomes

Go to Github and follow the instructions:

https://github.com/karkman/MMB-127_Genomics

(**Day 4:** Genome assembly)